

A ONE-DIMENSIONAL COHERENT DOMAIN WITH NON-PRÜFER INTEGRAL CLOSURE

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ABSTRACT. In a question recorded in 1978, Vasconcelos asked whether the integral closure of a one-dimensional coherent local domain in its fraction field is Prüfer. We construct such a domain whose integral closure is not Prüfer, giving a negative answer.

1. INTRODUCTION

All rings are commutative with identity. A domain D is *coherent* if every finitely generated ideal is finitely presented, and *Prüfer* if every nonzero finitely generated ideal is invertible. Equivalently, coherence requires finite generation of

$$\ker(D^r \xrightarrow{(a_1, \dots, a_r)} D) \quad (r \geq 1, a_i \in D).$$

Thus coherence controls relations among finite sets of generators, whereas Noetherianity requires every ideal to be finitely generated. Both Noetherian and Prüfer domains are coherent [8, §6].

The *integral closure* of D in $\text{Frac } D$ is

$$D' = \{x \in \text{Frac } D : f(x) = 0 \text{ for some monic } f \in D[T]\}.$$

The domain D is *integrally closed* (or *normal*) if $D' = D$.

Two one-dimensional results explain the motivation:

$$\begin{aligned} D \text{ Noetherian, } \dim D = 1 &\implies D' \text{ Dedekind,} \\ D \text{ coherent, } D = D', \dim D = 1 &\implies D \text{ Prüfer.} \end{aligned} \tag{1.1}$$

The first is the Krull–Akizuki theorem [21, Tag 09IG]; the second appears in [15, Corollary 11]. Since Dedekind domains are Prüfer, these implications suggest replacing Noetherianity by coherence: must every one-dimensional coherent domain have Prüfer integral closure? The issue is whether the finite relations in D impose enough control on its integral elements. Because $\dim D' = \dim D$ and Prüfer domains are coherent, the question is equivalently whether D' is coherent whenever D is one-dimensional and coherent.

The local form was already recorded by Papick in 1978 [14, §3]. Glaz–Vasconcelos formulated the assertion as Conjecture C₃ in their study of flat ideals [10, p. 223]. It subsequently appeared as Problem 65 in Chapman–Glaz’s 2000 collection [5] and Problem 3 in the 2014 collection of Cahen–Fontana–Frisch–Glaz [3]. The survey of Geroldinger–Kim–Loper still lists it as Problem 5 in the version dated 28 February 2026 [8, §6].

The positive results impose extra control on the base, its integral closure, or its overrings. Examples include very finite extensions, finite generation of the integral closure, coherence of overrings, and stability of ideals; Section 4 gives the precise hypotheses and their relationships. A complementary result shows the severity of a possible failure: in a one-dimensional normal local domain that is not Prüfer, the intersection of any two incomparable principal ideals is not finitely generated [11, Remark 2.2]. The 2026 survey highlights this obstruction and the scarcity of known coherent examples outside the Noetherian and Prüfer classes [8, §6].

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We give a negative answer, already for a local domain.

Theorem 1.1. *There exists a one-dimensional coherent local domain whose integral closure is not Prüfer.*

The construction separates two kinds of bounds. In the plane-curve domains $A_n = \mathbb{Q}[[t^n, t^{n+1}]]$, relations on r elements require at most r generators, independently of n , while the conductor scale grows quadratically. The first bound makes $A = \prod_{\mathcal{U}} A_n$ coherent. At the second scale, two prime order cuts $P \subsetneq Q$ satisfy

$$aP = P \quad (a \notin P), \quad R = (A/P)_{Q/P}, \quad \text{Spec } R = \{0, \mathfrak{m}\}.$$

The divisibility identity lifts relations through $A \rightarrow A/P$; localization then gives the coherent one-dimensional domain R of Section 2.

In Section 3, irrationally spaced exponents yield a fraction $z \in \text{Frac } R$ for which

$$R[z]/\mathfrak{m}R[z] \cong (R/\mathfrak{m})[T].$$

Thus $\dim R[z] \geq 2$. A one-dimensional Prüfer integral closure would force every overring of R to have dimension at most one, giving the contradiction. The proof detects the failure through a single overring and does not require an explicit description of the integral closure.

2. THE CONSTRUCTION

Fix a nonprincipal ultrafilter $\mathcal{U}$ on $\{2, 3, \dots\}$ and set

$$A_n = \mathbb{Q}[[t^n, t^{n+1}]] \subseteq \mathbb{Q}[[t]], \quad A = \prod_{\mathcal{U}} A_n.$$

Here A_n is the image of $\mathbb{Q}[[X, Y]]$ under $(X, Y) \mapsto (t^n, t^{n+1})$. Write “almost everywhere” for a set belonging to $\mathcal{U}$; thus $[a_n] = [b_n]$ if $a_n = b_n$ almost everywhere. The ultrafilter alternative makes A a domain. Let ν_n be the t -adic order on $\mathbb{Q}((t))$, with $\nu_n(0) = \infty$. We use

$$\nu_n(ab) = \nu_n(a) + \nu_n(b), \quad \nu_n\left(\sum_i a_i\right) \geq \min_i \nu_n(a_i),$$

with equality in the second formula when the minimum is unique. Define two order cuts:

$$P = \{[a_n] : (\forall C \in \mathbb{Z}_{\geq 1}) \{n : \nu_n(a_n) > Cn^2\} \in \mathcal{U}\}, \quad (2.1)$$

$$Q = \{[a_n] : (\exists \epsilon > 0) \{n : \nu_n(a_n) > \epsilon n^2\} \in \mathcal{U}\}. \quad (2.2)$$

Each quantified bound may hold on a different set in $\mathcal{U}$; only finite intersections are used below.

Lemma 2.1. *For $H_n = \langle n, n+1 \rangle$ and $e = qn + r \geq 0$, $0 \leq r < n$, $e \in H_n \iff q \geq r$. Consequently*

$$A_n = \bigoplus_{r=0}^{n-1} t^{r(n+1)} \mathbb{Q}[[t^n]] \cong \mathbb{Q}[[X, Y]]/(Y^n - X^{n+1}), \quad t^{n^2} \mathbb{Q}[[t]] \subseteq A_n. \quad (2.3)$$

Proof. If $q \geq r$, then $e = (q - r)n + r(n + 1)$. Conversely, from $e = an + b(n + 1)$ and $b = kn + r$ one obtains $q = a + b + k \geq r$. Grouping series supported on H_n by their exponents modulo n gives the direct sum. The conductor inclusion follows from $e \geq n(n - 1) \Rightarrow q \geq n - 1 \geq r$.

For the presentation, set $f = Y^n - X^{n+1}$. Modulo f , every $F = \sum_{b \geq 0} f_b(X)Y^b$ has the remainder

$$\sum_{r < n} g_r(X)Y^r, \quad g_r(X) = \sum_{k \geq 0} f_{kn+r}(X)X^{k(n+1)}.$$

For $k \geq 1$, every monomial of $(Y^{kn} - X^{k(n+1)})/f$ has total degree at least $n(k - 1)$, so the quotient series and displayed sums converge formally. Substitution sends distinct remainder terms to distinct exponent classes modulo n . A zero image therefore forces every $g_r = 0$, proving that the kernel is (f) . $\square$

Lemma 2.2. *For every $n \geq 2$ and $r \geq 1$, the relation module of any r elements of A_n has at most r generators. Hence A is coherent.*

Proof. Write $\text{Rel}_D(a_i) = \ker((a_i) : D^r \rightarrow D)$. Set $S = \mathbb{Q}[[X, Y]]$, $f = Y^n - X^{n+1}$, and $I = (a_1, \dots, a_r)A_n$. If $I = 0$, the relation module is A_n^r . Otherwise let $K = \ker(S^r \rightarrow I)$. We use standard power-series, DVR, and Nakayama facts [21, Tags 0306, 0AUW, 00DV]. Since X acts on I as the nonzero element t^n ,

$$K \cap XS^r = XK, \quad K/XK \hookrightarrow \mathbb{Q}[[Y]]^r.$$

This submodule is free of rank $\ell \leq r$. Lifting a basis gives $\psi : S^\ell \rightarrow K$, surjective by Noetherianity and Nakayama. For $L = \ker \psi$, the isomorphism modulo X gives $L \subseteq XS^\ell$. If $Xw \in L$, injectivity of X on $K \subseteq S^r$ gives $w \in L$. Thus $L = XL$, so Noetherianity and Nakayama give $L = 0$. Since $f \neq 0$ annihilates I , tensoring with $\text{Frac } S$ gives $\ell = r$. Therefore

$$\text{Rel}_{A_n}(a_1, \dots, a_r) = K/fS^r$$

is generated by the images of an S -basis of K .

For a row of length r over A , choose r relation generators in each coordinate, padding by zeros. Every relation over A holds coordinatewise on a set in $\mathcal{U}$, where it is a linear combination of these generators; extend the coefficients by zero elsewhere. Their classes therefore generate the relation module over A . A ring is coherent exactly when every finite row has a finitely generated relation module [21, §10.90]. Hence A is coherent. $\square$

Proposition 2.3. *The ideals $P \subsetneq Q$ are prime. The ring*

$$R = (A/P)_{Q/P}$$

is a one-dimensional coherent local domain, with maximal ideal $\mathfrak{m} = (Q/P)R$. Writing bars for images in R , the element $u = [t^{n^2}]$ satisfies $0 \neq \bar{u} \in \mathfrak{m}$.

Proof. The order inequalities make P, Q proper ideals. If $a, b \notin P$, choose respective order bounds Cn^2, Dn^2 almost everywhere; then $\nu_n(a_nb_n) \leq (C+D)n^2$ almost everywhere, so $ab \notin P$. If $a, b \notin Q$, for each fixed $\epsilon > 0$ both orders are at most $\epsilon n^2/2$ almost everywhere, giving $ab \notin Q$. Thus P, Q are prime, with $P \subsetneq Q$ witnessed by $u \in Q \setminus P$ and (2.3).

For $a \notin P$ we claim that

$$aP = P. \tag{2.4}$$

Choose $C \geq 1$ with $\nu_n(a_n) \leq Cn^2$ almost everywhere. For $p \in P$, define $b_n = p_n/a_n$ where this bound holds and $\nu_n(p_n) > (C+1)n^2$, and set $b_n = 0$ elsewhere. The conductor inclusion puts b_n in A_n , and $ab = p$. For every fixed integer $D \geq 1$, intersecting the bound on a_n with $\nu_n(p_n) > (C+D)n^2$ gives $\nu_n(b_n) > Dn^2$ almost everywhere. Thus this single b lies in P , proving (2.4).

For a nonzero finite row over A/P , choose a lift (a_i) with $a_j \notin P$. Any lifted relation (b_i) has error $p = \sum_i a_i b_i \in P$. By (2.4), write $p = a_j c$ with $c \in P$; replacing b_j by $b_j - c$ gives an exact relation with the same reduction. Hence

$$\text{Rel}_A(a_i) \twoheadrightarrow \text{Rel}_{A/P}(a_i + P).$$

The relation criterion gives coherence of A/P (the zero row is immediate). Exactness of localization gives coherence of R . Primality makes R a local domain with the stated maximal ideal.

For $a \in Q \setminus P$, we next show

$$\sqrt{aA} = Q. \tag{2.5}$$

Choose C as above. For $b \in Q$, choose $\epsilon > 0$ with $\nu_n(b_n) > \epsilon n^2$ almost everywhere, and then an integer $N \geq 1$ with $N\epsilon > C+1$. Then b_n^N/a_n has order greater than n^2 on the common set. The conductor inclusion, with zero coordinates elsewhere, yields $b^N \in aA$. The reverse inclusion

follows from primality of Q . If $P \subsetneq \mathfrak{p} \subseteq Q$ is prime, choose $a \in \mathfrak{p} \setminus P$; (2.5) forces $\mathfrak{p} = Q$. Prime correspondence gives $\text{Spec } R = \{0, \mathfrak{m}\}$, and $0 \neq \bar{u} \in \mathfrak{m}$ proves $\dim R = 1$. $\square$

3. THE OBSTRUCTION TO PRÜFER INTEGRAL CLOSURE

Set $F = \text{Frac } R$ and

$$\alpha = \sqrt{2} - 1, \quad d_n = \lfloor \alpha n \rfloor, \quad v = [t^{n^2+d_n}], \quad z = \bar{v}/\bar{u} \in F.$$

The conductor gives $u, v \in A$, and their orders are at most $2n^2$, so $u, v \notin P$. Thus $z \in F$, although t^{d_n} need not lie in A_n . Irrationality will separate the low-order terms of a relation for z .

Lemma 3.1. *Every polynomial in $R[T]$ vanishing at z has all coefficients in $\mathfrak{m}$. Consequently*

$$R[z]/\mathfrak{m}R[z] \cong (R/\mathfrak{m})[T], \quad \dim R[z] \geq 2.$$

Proof. Suppose $\sum_{i=0}^d c_i z^i = 0$ and some $c_i \notin \mathfrak{m}$; necessarily $d \geq 1$. Clear a common denominator to write $c_i = \bar{b}_i/\bar{s}$, with $b_i \in A$ and $s \notin Q$. Then $J = \{i : b_i \notin Q\} \neq \emptyset$ and, since $A/P \hookrightarrow F$,

$$h = \sum_{i=0}^d b_i v^i u^{d-i} \in P. \quad (3.1)$$

For this fixed degree, irrationality gives

$$\delta = \min_{1 \leq k \leq d} \text{dist}(k\alpha, \mathbb{Z}) > 0.$$

Choose a common $\epsilon > 0$ with $\nu_n(b_{i,n}) > \epsilon n^2$ almost everywhere for all $i \notin J$ (take $\epsilon = 1$ if there are none), and fix $0 < \eta < \min\{\epsilon/4, \delta/4, 1/4\}$. On a common set $E \in \mathcal{U}$ these lower bounds hold, $\nu_n(b_{i,n}) \leq \eta n^2$ for every $i \in J$, and $d/n < \min\{\delta/4, \eta\}$.

For $n \in E$ and $i \in J$, write the leading exponent as $e_{i,n} = q_{i,n}n + r_{i,n}$, $0 \leq r_{i,n} < n$. The semigroup criterion gives

$$0 \leq r_{i,n}/n \leq q_{i,n}/n \leq e_{i,n}/n^2 \leq \eta.$$

The shifted orders $e_{i,n} + id_n$ are distinct for $i \in J$. Otherwise equality for $i \neq j$, using $q_{i,n} - q_{j,n} \in \mathbb{Z}$, would give

$$\delta \leq \text{dist}((i-j)\alpha, \mathbb{Z}) \leq \frac{|r_{i,n} - r_{j,n}|}{n} + |i-j| \left| \alpha - \frac{d_n}{n} \right| < \eta + \frac{d}{n} < \frac{\delta}{2}.$$

For $i \in J$, the order of $b_{i,n}t^{id_n}$ is less than $2\eta n^2 < \epsilon n^2$; for $i \notin J$ it exceeds ϵn^2 . Thus $g_n = \sum_i b_{i,n}t^{id_n}$ has a unique summand of least order, and $\nu_n(g_n) < 2\eta n^2 < n^2$. Since $h_n = t^{dn^2}g_n$, this gives $\nu_n(h_n) < (d+1)n^2$ on E , contradicting (3.1).

The evaluation kernel lies in $\mathfrak{m}R[T]$, giving the quotient isomorphism. The primes $(0) \subsetneq (T)$ of $(R/\mathfrak{m})[T]$ yield the prime chain

$$0 \subsetneq \mathfrak{m}R[z] \subsetneq \mathfrak{m}R[z] + zR[z].$$

The first inclusion is strict since $0 \neq \bar{u} \in \mathfrak{m}R[z]$; hence $\dim R[z] \geq 2$. $\square$

Lemma 3.2. *Every overring of a one-dimensional Prüfer domain has dimension at most one.*

Proof. Let $D \subseteq E \subseteq \text{Frac } D$, with D Prüfer. For $\mathfrak{q} \in \text{Spec } E$ and $\mathfrak{p} = \mathfrak{q} \cap D$, the local ring $E_{\mathfrak{q}}$ dominates $D_{\mathfrak{p}}$. The latter is a valuation domain: its nonzero finitely generated ideals are invertible, hence principal, and a local Bézout domain is a valuation domain. If $x \in E_{\mathfrak{q}} \setminus D_{\mathfrak{p}}$, then $x^{-1} \in \mathfrak{p}D_{\mathfrak{p}} \subseteq \mathfrak{q}E_{\mathfrak{q}}$, contradicting $xx^{-1} = 1$. Thus $E_{\mathfrak{q}} = D_{\mathfrak{p}}$, and $\dim E = \sup_{\mathfrak{q}} \dim E_{\mathfrak{q}} \leq \dim D$. $\square$

Proof of Theorem 1.1. By Proposition 2.3, R is a one-dimensional coherent local domain. Let B be its integral closure in F ; then $\text{Frac } B = F$. Integral extensions preserve dimension [21, Tag 00OK], so $\dim B = 1$. If B were Prüfer, Lemma 3.2 would give $\dim B[z] \leq 1$. Since $B[z]$ is generated over $R[z]$ by elements integral over R , the extension $R[z] \subseteq B[z]$ is integral. Dimension preservation and Lemma 3.1 give

$$2 \leq \dim R[z] = \dim B[z] \leq 1,$$

a contradiction. $\square$

4. POSITIVE CRITERIA AND THE OBSTRUCTION

The positive results concern finiteness in the base ring, preservation of coherence in extensions, and ideal-theoretic conditions on the integral closure. We compare these approaches and identify the obstruction supplied by our example. Throughout, D is a one-dimensional coherent domain, $K = \text{Frac } D$, and D' is its integral closure in K , unless weaker hypotheses are stated. An overring lies between D and K .

4.1. Finiteness and extension properties. Krull–Akizuki gives more than Prüfer integral closure in the Noetherian case: D' is Dedekind. Even the Mori and H conditions force a one-dimensional coherent domain to be Noetherian [10, p. 225]. Weakening Noetherianity in these directions therefore does not reach the general problem. A different approach asks whether coherence passes to D' . In dimension one, it is enough that D' be locally finite-conductor; D' module-finite over D is one sufficient condition [17, Thm. 3.7, Cor. 3.9]. Finite conductor means that $aD \cap bD$ is finitely generated for all $a, b \in D$; coherence implies this by applying the relation criterion to $(a, -b)$.

For a nonzero ideal J of D , put

$$J^{-1} = (D : J), \quad J_v = (J^{-1})^{-1}, \quad J_t = \bigcup_{\substack{L \subseteq J \\ L \text{ finitely generated}}} L_v \quad (J \neq 0),$$

with $0_v = 0$. Divisorial means $J = J_v$; Mori means ACC on divisorial ideals. The H condition [9, §3] is

$$J^{-1} = D \implies \exists L \subseteq J \text{ finitely generated with } L^{-1} = D.$$

A DW domain has every maximal ideal a t -ideal; this is automatic in dimension one [17, Prop. 2.9]. Thus the normal DW finite-conductor criterion applies to D' whenever its localizations are finite-conductor. The missing inheritance property is precise: finiteness of relations in D need not yield finiteness of principal-ideal intersections in D' .

Every row of Table 1 implies that D' is Prüfer; stronger conclusions and changes to the standing assumptions are indicated. A *very finite* extension has every intermediate ring, including the endpoint, module-finite over the base; a *coherent pair* has every intermediate ring coherent. These hypotheses control intermediate rings, whereas finite generation of an endpoint alone does not.

Table 1: Additional hypotheses ensuring Prüfer integral closure.
Each alternative includes its stated conjunction.

Additional hypothesis	Result and scope
Noetherian; Mori; H -domain	D Noetherian, D' Dedekind [21, Tag 09IG]; [10, p. 225].
Integrally closed; regular; finite weak or global dimension; Prüfer v -multiplication domain; v -domain	D Prüfer [15, Cor. 11]; [20, Prop. 3.1]. Regular means that every finitely generated ideal has finite projective dimension.

Continued on the next page

Table 1 continued

Additional hypothesis	Result and scope
D' module-finite over D ; D' locally finite-conductor	[17, Thm. 3.7, Cor. 3.9]. The second includes locally coherent or coherent D' and needs no coherence assumption on D .
Each nonnormal $D_{\mathfrak{p}}$ has a proper very finite overring	Each such $D_{\mathfrak{p}}$ is Noetherian [14, Thm. 15, Cor. 16].
D local, with a proper very finite domain extension or a proper integral coherent pair	D Noetherian [12, Thm. 2]; [15, Rem. 9]. The extension need not lie in K .
Every proper overring coherent; every local overring coherent	[16, Thm. 1, Rem. 2]. Neither dimension one nor coherence of D is required.
Locally a pseudo-valuation domain	[6, Cor. 3.6] and localization. A pseudo-valuation domain has a valuation overring sharing its maximal ideal.
A canonical module and finitely generated uppers to zero	[10, Prop. 4.14]. Coherence suffices; dimension one is unnecessary. Evaluation and endomorphism formulations are discussed below.
Finitely stable; stable; quasi-stable	[18, Prop. 3.4, Cor. 3.5]. No dimension or coherence hypothesis is needed.
Clifford regular	D' Prüfer of finite character [2, Prop. 3.4, Cor. 4.8]; no standing assumptions needed.
Totally divisorial; Warfield	D stable and, in dimension one, Noetherian [13, pp. 263–264, Lemmas 3.2–3.3].
Finite expansion over a field k : integral over a finitely generated k -subalgebra	Apply [19, Thm. 3.5] to D' . Integrality preserves dimension and finite expansion; coherence is unnecessary.

Since B is not Prüfer, none of the full sufficient hypotheses in the table holds for our example. In particular, R is non-Noetherian and nonnormal, B is not module-finite, and some $B_{\mathfrak{n}}$ is not finite-conductor. This last failure is sharp: in a one-dimensional normal local domain, intersections of incomparable principal ideals are not finitely generated [11, Rem. 2.2].

4.2. Stability and polynomial relations. An ideal $J \neq 0$ is *stable* if invertible over $(J : J)$. Finite stability tests finitely generated ideals, stability tests all nonzero ideals, and quasi-stability replaces invertibility by flatness. These give a route to Prüfer integral closure through endomorphism rings. Clifford regularity means that the semigroup of nonzero fractional ideals modulo principal ideals is von Neumann regular; it implies finite stability. Total divisoriality requires every ideal of every overring to be divisorial; Warfield duality implies this condition [2, 13]. Divisoriality of finitely generated ideals already forces a domain to be Prüfer when normality is added [20, Cor. 3.2].

Polynomial relations connect these criteria to the construction. An *upper to zero* is a nonzero prime of $D[T]$ contracting to zero. Write $c(f)$ for the ideal of coefficients of f ; f is primitive if

$c(f) = D$. For arbitrary domains, the characterizations collected in [4, Thm. 1.1] give

$$\begin{aligned} D' \text{ Prüfer} &\iff D \text{ quasi-Prüfer} \\ &\iff \text{every upper to zero contains a primitive polynomial} \\ &\iff \text{every } x \in K \text{ satisfies a primitive equation over } D \\ &\iff \text{every overring satisfies INC over } D. \end{aligned}$$

Here INC means that comparable primes have distinct contractions. Equivalent conditions are that every overring is quasi-Prüfer, or that every prime of the Nagata ring $D[T]_N$ is extended from D , where N consists of primitive polynomials. In dimension one, these are also exactly the UMT domains, in which every upper to zero is a maximal t -ideal; in arbitrary dimension, D' is Prüfer exactly when every overring is UMT [7, Thms. 1.1, 1.5, Cor. 3.11].

Our single evaluation upper displays these failures simultaneously. With $\mathcal{H} = \ker(R[T] \rightarrow R[z])$, $T \mapsto z$, Lemma 3.1 gives

$$0 \neq \bar{u}T - \bar{v} \in \mathcal{H} \subsetneq \mathfrak{m}[T], \quad R[z]/\mathfrak{m}R[z] \cong (R/\mathfrak{m})[T]. \quad (4.1)$$

Thus z has no primitive equation, and $R[z]$ has comparable primes above $\mathfrak{m}$. Its fiber has a transcendental residue-field extension, the obstruction to residual algebraicity studied in [1, Thm. 2.3, Cor. 2.8]. Since $\mathfrak{m}$ is a t -ideal, this upper excludes UMT by [7, Thm. 1.1]. It also yields

$$0 \subsetneq \mathcal{H} \subsetneq \mathfrak{m}[T] \subsetneq \mathfrak{m}[T] + TR[T], \quad \dim R[T] \geq 3.$$

For comparison, a one-dimensional domain with $\dim D[T] = 2$ is quasi-Prüfer: failure produces an upper inside an extended maximal ideal and hence a chain of length three. In particular, R is not Jaffard, since that property requires $\dim R[T] = \dim R + 1$.

Glaz–Vasconcelos also link finite generation of relations to stabilization. For a coherent D , $J = (a, b)$ with $b \neq 0$, and $\mathcal{H}_{a/b} = \ker(D[T] \rightarrow D[a/b])$, their Propositions 4.10–4.11 give [10]

$$\begin{aligned} D[JT] \text{ coherent} &\implies (J^n : J^n) \text{ eventually constant} \\ &\iff \mathcal{H}_{a/b} \text{ finitely generated.} \end{aligned}$$

A canonical module supplies duality on finitely generated torsion-free modules. With this hypothesis, finite generation of all evaluation uppers suffices in the proof of their Proposition 4.14. Their Proposition 4.13 gives the exact ideal-theoretic criterion

$$D' \text{ Prüfer} \iff \forall J = (a, b) \neq 0 \exists n \geq 1 : J^n \text{ invertible over } (J^n : J^n).$$

A failed conjunction does not exclude each factor: the present argument leaves existence of a canonical module for R , finite generation of $\mathcal{H}$, and coherence of $R[T]$ and $R[z]$ undetermined. The canonical-ideal claim reported as a private communication in [15, p. 336] is not used as a theorem.

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REFERENCES

- [1] Ahmed Ayache and Ali Jaballah. Residually algebraic pairs of rings. *Mathematische Zeitschrift*, 225:49–65, 1997.
- [2] Silvana Bazzoni. Finite character of finitely stable domains. *Journal of Pure and Applied Algebra*, 215(6):1127–1132, 2011.

- [3] Paul-Jean Cahen, Marco Fontana, Sophie Frisch, and Sarah Glaz. Open problems in commutative ring theory. In Marco Fontana, Sophie Frisch, and Sarah Glaz, editors, *Commutative Algebra: Recent Advances in Commutative Rings, Integer-Valued Polynomials, and Polynomial Functions*, pages 353–375. Springer, New York, 2014. doi:10.1007/978-1-4939-0925-4_20.
- [4] Gyu Whan Chang and Marco Fontana. Uppers to zero in polynomial rings and Prüfer-like domains. *Communications in Algebra*, 37(1):164–192, 2009.
- [5] Scott T. Chapman and Sarah Glaz. One hundred problems in commutative ring theory. In Scott T. Chapman and Sarah Glaz, editors, *Non-Noetherian Commutative Ring Theory*, volume 520 of *Mathematics and Its Applications*, pages 459–476. Kluwer Academic Publishers, Dordrecht, 2000. doi:10.1007/978-1-4757-3180-4_21.
- [6] David E. Dobbs. On the weak global dimension of pseudo-valuation domains. *Canadian Mathematical Bulletin*, 21(2):159–164, 1978.
- [7] Marco Fontana, Stefania Gabelli, and Evan Houston. UMT-domains and domains with Prüfer integral closure. *Communications in Algebra*, 26(4):1017–1039, 1998.
- [8] Alfred Geroldinger, Hwankoo Kim, and K. Alan Loper. On long-term problems in multiplicative ideal theory and factorization theory. [arXiv:2502.21020v3 \[math.AC\]](https://arxiv.org/abs/2502.21020v3), 2026. Version 3, 28 February 2026.
- [9] Sarah Glaz and Wolmer V. Vasconcelos. Flat ideals II. *Manuscripta Mathematica*, 22(4):325–341, 1977.
- [10] Sarah Glaz and Wolmer V. Vasconcelos. Flat ideals III. *Communications in Algebra*, 12(1–2):199–227, 1984.
- [11] Lorenzo Guerrieri and K. Alan Loper. On the integral domains characterized by a Bezout property on intersections of principal ideals. *Journal of Algebra*, 586:208–231, 2021.
- [12] James A. Huckaba and Ira J. Papick. Remarks on module-finite pairs. *Proceedings of the Edinburgh Mathematical Society*, 24(2):137–139, 1981.
- [13] Bruce Olberding. Stability, duality, 2-generated ideals and a canonical decomposition of modules. *Rendiconti del Seminario Matematico della Università di Padova*, 106:261–290, 2001.
- [14] Ira J. Papick. Finite type extensions and coherence. *Pacific Journal of Mathematics*, 78(1):161–172, 1978.
- [15] Ira J. Papick. Coherent overrings. *Canadian Mathematical Bulletin*, 22(3):331–337, 1979.
- [16] Ira J. Papick. A note on proper overrings. *Commentarii Mathematici Universitatis Sancti Pauli*, 28(2):137–140, 1979.
- [17] Giampaolo Picozza and Francesca Tartarone. When the semistar operation $\widetilde{\star}$ is the identity. *Communications in Algebra*, 36(5):1954–1975, 2008.
- [18] Giampaolo Picozza and Francesca Tartarone. Flat ideals and stability in integral domains. *Journal of Algebra*, 324(8):1790–1802, 2010.
- [19] Matthias Johann Steiner. Schemes of finite expansion and universally closed curves. *Beiträge zur Algebra und Geometrie / Contributions to Algebra and Geometry*, 67:371–393, 2026.
- [20] Mohammed Tamekkante, Refat Abdelmawla Khaled Assaad, and El Mehdi Bouba. Note on the DW rings. *International Electronic Journal of Algebra*, 25:43–54, 2019.
- [21] The Stacks Project Authors. The Stacks project. stacks.math.columbia.edu, 2026. Accessed 14 September 2026.

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